

VM Network Configuration

VM / Device	IP Address	Role / Notes	Network Interface
Wazuh	192.168.1.100	Security engine (ML scoring, alert correlation)	LAN / internal
Ubuntu	192.168.1.101	ELK stack (logging, visualization)	LAN / internal
Windows 10	192.168.1.102	Target system for attack simulation	LAN / internal
pfSense	192.168.1.1	Firewall / automated response	LAN
	10.0.2.15	WAN interface (DHCP from host NAT)	WAN
	192.168.56.100	OPT1 – host-only (optional / isolated tests)	OPT1
	192.168.57.1	OPT2 – internal network	OPT2
Kali Linux	192.168.57.10	Attack simulation / pentesting	OPT2



Details

Snapshots

Logs

Resource Use

File Manager

General

System

Base Memory: 808 MB

Boot Order: Floppy, Optical, Hard Disk

Acceleration: Nested Paging

Display

Video Memory: 16 MB

Graphics Controller: VMSVGA

Remote Desktop Server: Disabled

Recording: Disabled

Storage

Controller: IDE

IDE Primary Device 0: pfSense2.6.0.vdi (Normal, 16.00 GB)

IDE Secondary Device 0: [Optical Drive] Empty

Audio

Host Driver: Default

Controller: ICH AC97

Network

Adapter 1: Intel PRO/1000 MT Desktop (NAT)

Adapter 2: Intel PRO/1000 MT Desktop (Internal Network, 'LABnet')

Adapter 3: Intel PRO/1000 MT Desktop (Host-only Adapter, 'VirtualBox Host-Only Ethernet Adapter')

Adapter 4: Intel PRO/1000 MT Desktop (Internal Network, 'KALInet')

USB

USB Controller: OHCI, EHCI

Device Filters: 0 (0 active)

Shared folders

None

Figure 1 VM configurations for pfSense setup

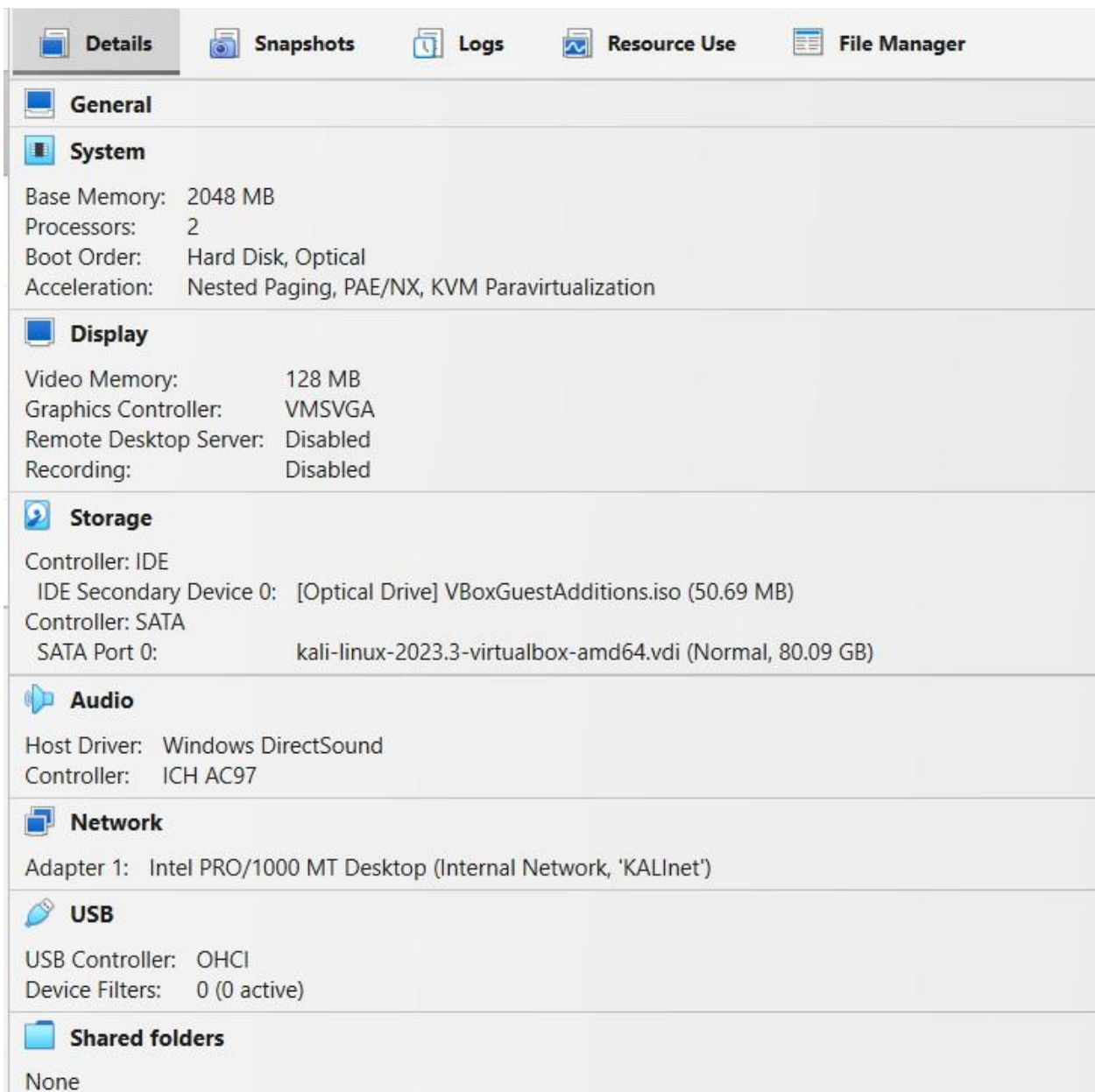


Figure 2 VM configurations for Kali Linux setup

Details

Snapshots

Logs

Resource Use

File Manager

General

Name:

windows

Operating System:

Windows 10 (64-bit)

System

Base Memory:

4096 MB

Processors:

2

Boot Order:

Hard Disk, Optical, Floppy

Acceleration:

Nested Paging, Hyper-V Paravirtualization

Display

Video Memory:

128 MB

Graphics Controller:

VBoxSVGA

Remote Desktop Server:

Disabled

Recording:

Disabled

Storage

Controller:

SATA

SATA Port 0:

windows_vdi (Normal, 50.00 GB)

SATA Port 1:

[Optical Drive] Win10_22H2_English_x64v1.iso (5.72 GB)

Controller:

Floppy

Floppy Device 0:

Unattended-ccff81fa-3d8d-4b2c-99ad-9d9de51d3e74-aux-floppy.img (1.41 MB)

Audio

Host Driver:

Default

Controller:

Intel HD Audio

Network

Adapter 1:

Intel PRO/1000 MT Desktop (Internal Network, 'LABnet')

USB

USB Controller:

xHCI

Device Filters:

0 (0 active)

Figure 3 VM configurations for Windows setup

 Details

 Snapshots

 Logs

 Resource Use

 File Manager

 **General**

Name: ubuntu_server
Operating System: Ubuntu (64-bit)

 **System**

Base Memory: 5085 MB
Processors: 2
Boot Order: Hard Disk, Optical, Floppy
Acceleration: Nested Paging, KVM Paravirtualization

 **Display**

Video Memory: 16 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled

 **Storage**

Controller: IDE
IDE Primary Device 0: [Optical Drive] VBoxGuestAdditions.iso (50.69 MB)
Controller: SATA
SATA Port 0: ubuntu_server.vdi (Normal, 30.14 GB)

 **Audio**

Host Driver: Default
Controller: ICH AC97

 **Network**

Adapter 1: Intel PRO/1000 MT Desktop (Host-only Adapter, "VirtualBox Host-Only Ethernet Adapter")
Adapter 2: Intel PRO/1000 MT Desktop (Internal Network, "LABnet")

 **USB**

USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)

Figure 4 VM configurations for Ubuntu setup

Details
Snapshots
Logs
Resource Use
File Manager

General

Name: Wazuh v4.14.1 OVA
Operating System: Linux 2.6 / 3.x / 4.x / 5.x (64-bit)

System

Base Memory: 4096 MB
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, KVM Paravirtualization

Display

Video Memory: 16 MB
Graphics Controller: VBoxVGA
Remote Desktop Server: Disabled
Recording: Disabled

Storage

Controller: IDE
IDE Secondary Device 0: wazuh-4.14.1-disk-1.vdi (Normal, 25.00 GB)
Controller: Floppy
Floppy Device 0: Empty

Audio

Host Driver: Default
Controller: ICH AC97

Network

Adapter 1: Intel PRO/1000 MT Server (Internal Network, 'LABnet')

Figure 5 VM configurations for Wazuh setup

```

*** Welcome to pfSense 2.6.0-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 10.0.2.15/24
LAN (lan)      -> em1      -> v4: 192.168.1.1/24
OPT1 (opt1)    -> em2      -> v4: 192.168.56.100/24
OPT2 (opt2)    -> em3      -> v4: 192.168.57.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Disable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

```

Figure 6 pfSense CLI dashboard showing network interface statuses and management options

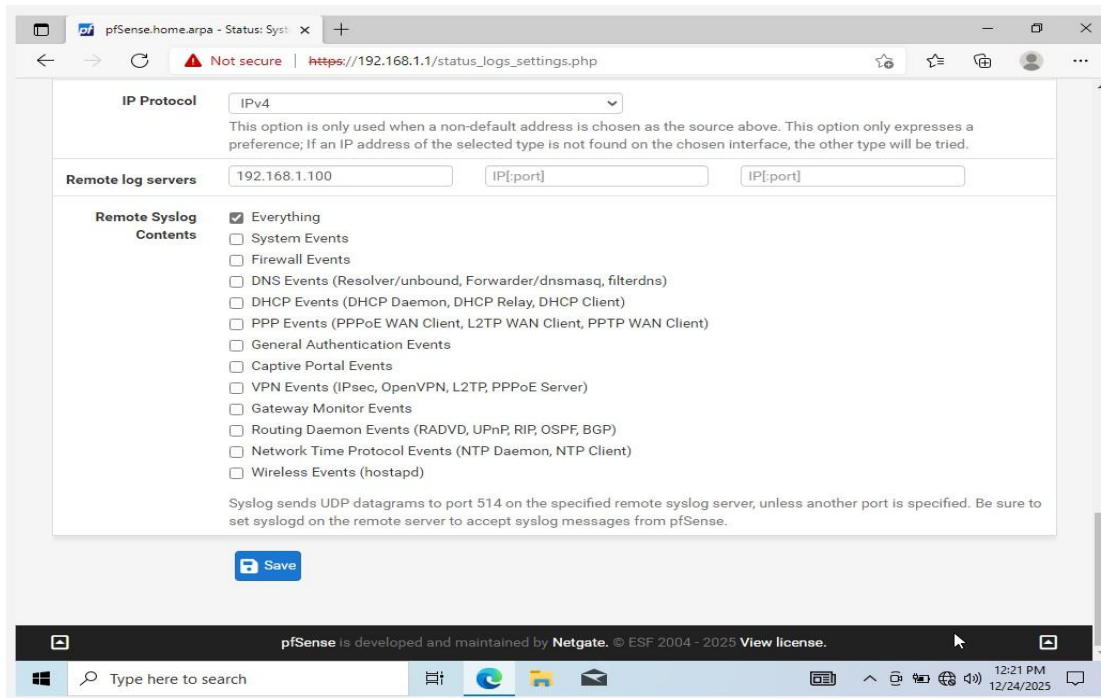


Figure 7 Configuring pfSense Remote Syslog Settings

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group d
  default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
  group default qlen 1000
    link/ether 08:00:27:cb:7e:f5 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.10/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::de3e:3809:bb98:9a32/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping 192.168.56.100
PING 192.168.56.100 (192.168.56.100) 56(84) bytes of data.
64 bytes from 192.168.56.100: icmp_seq=1 ttl=64 time=17.9 ms
64 bytes from 192.168.56.100: icmp_seq=2 ttl=64 time=1.26 ms
64 bytes from 192.168.56.100: icmp_seq=3 ttl=64 time=1.89 ms
^Z
zsh: suspended ping 192.168.56.100

(kali@kali)-[~]
$ ping 192.168.1.102
PING 192.168.1.102 (192.168.1.102) 56(84) bytes of data.
64 bytes from 192.168.1.102: icmp_seq=1 ttl=127 time=1050 ms
64 bytes from 192.168.1.102: icmp_seq=2 ttl=127 time=54.4 ms
64 bytes from 192.168.1.102: icmp_seq=3 ttl=127 time=10.7 ms
^Z
zsh: suspended ping 192.168.1.102
```

Figure 8 Network configuration verification through ping tests to ensure proper setup of the environment

```
wazuh-user@wazuh-server:~$ sudo systemctl stop wazuh
Killing wazuh-remoted...
Killing wazuh-syscheckd...
Killing wazuh-analysisd...
wazuh-malld not running...
Killing wazuh-execd...
Killing wazuh-db...
Killing wazuh-authd...
wazuh-agentlessd not running...
wazuh-integratord not running...
wazuh-dbd not running...
wazuh-csyslogd not running...
Killing wazuh-apid...
Process wazuh-authd couldn't be terminated. It will be killed.
Wazuh v4.14.3 Stopped
Starting Wazuh v4.14.3...
Started wazuh-apid...
Started wazuh-csyslogd...
Started wazuh-dbd...
2026/02/24 14:12:34 wazuh-integratord: INFO: Remote integrations not configured. Clean exit.
Started wazuh-integratord...
Started wazuh-agentlessd...
2026/02/24 14:12:34 wazuh-authd: WARNING: The <force_insert> tag is deprecated. Use <force> instead.
Started wazuh-authd...
Started wazuh-db...
Started wazuh-execd...
Started wazuh-analysisd...
2026/02/24 14:12:39 wazuh-syscheckd: INFO: (6678): No directory provided for syscheck to monitor.
2026/02/24 14:12:39 wazuh-syscheckd: INFO: (6001): File integrity monitoring disabled.
2026/02/24 14:12:39 rootcheck: INFO: Rootcheck disabled.
Started wazuh-syscheckd...
Started wazuh-remoted...
2026/02/24 14:12:41 wazuh-logcollector: INFO: (1905): No file configured to monitor.
Started wazuh-logcollector...
Started wazuh-monitor...
2026/02/24 14:12:43 wazuh-modulesd:router: INFO: Loaded router module.
2026/02/24 14:12:43 wazuh-modulesd:content_manager: INFO: Loaded content_manager module.
2026/02/24 14:12:43 wazuh-modulesd:inventory-harvester: INFO: Loaded Inventory harvester module.
Started wazuh-modulesd...
Completed.
[wazuh-user@wazuh-server ~]$ sudo ss -tlnp | grep 1515
tcp LISTEN 0      128          0.0.0.0:1515    0.0.0.0:*      users:(("wazuh-authd",pid=5766,fd=6))
```

Figure 9 Wazuh Manager starting with service initialization and log messages

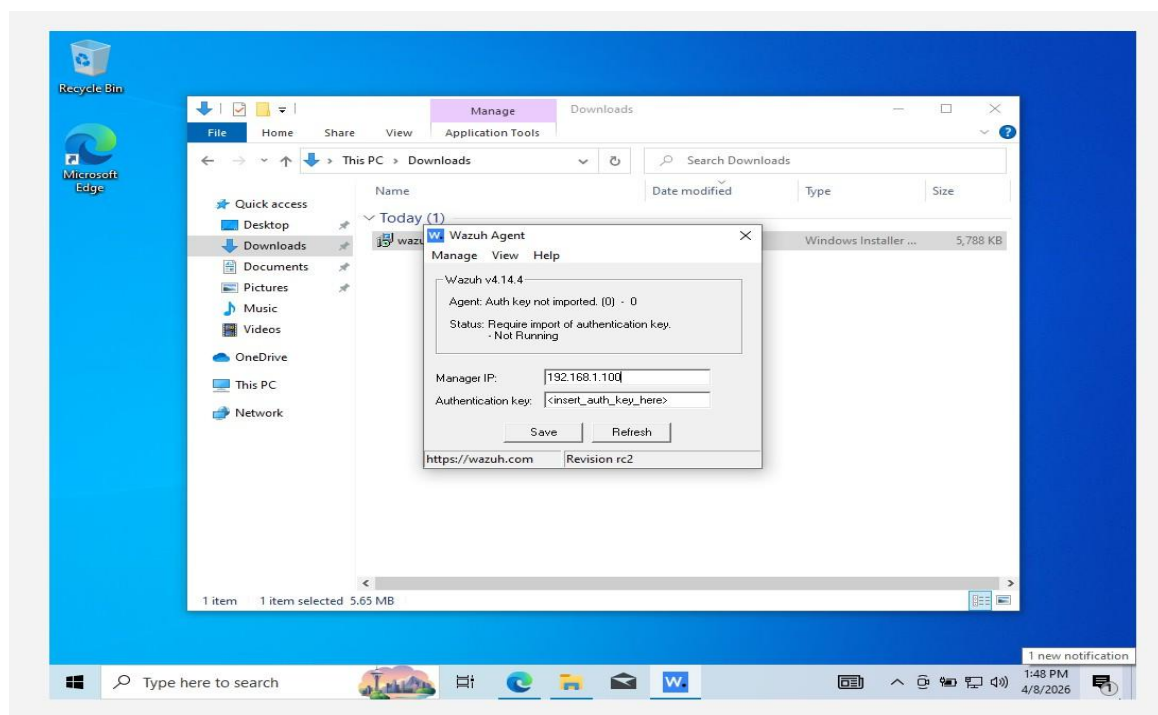


Figure 10 Wazuh Agent installation requiring authentication key.

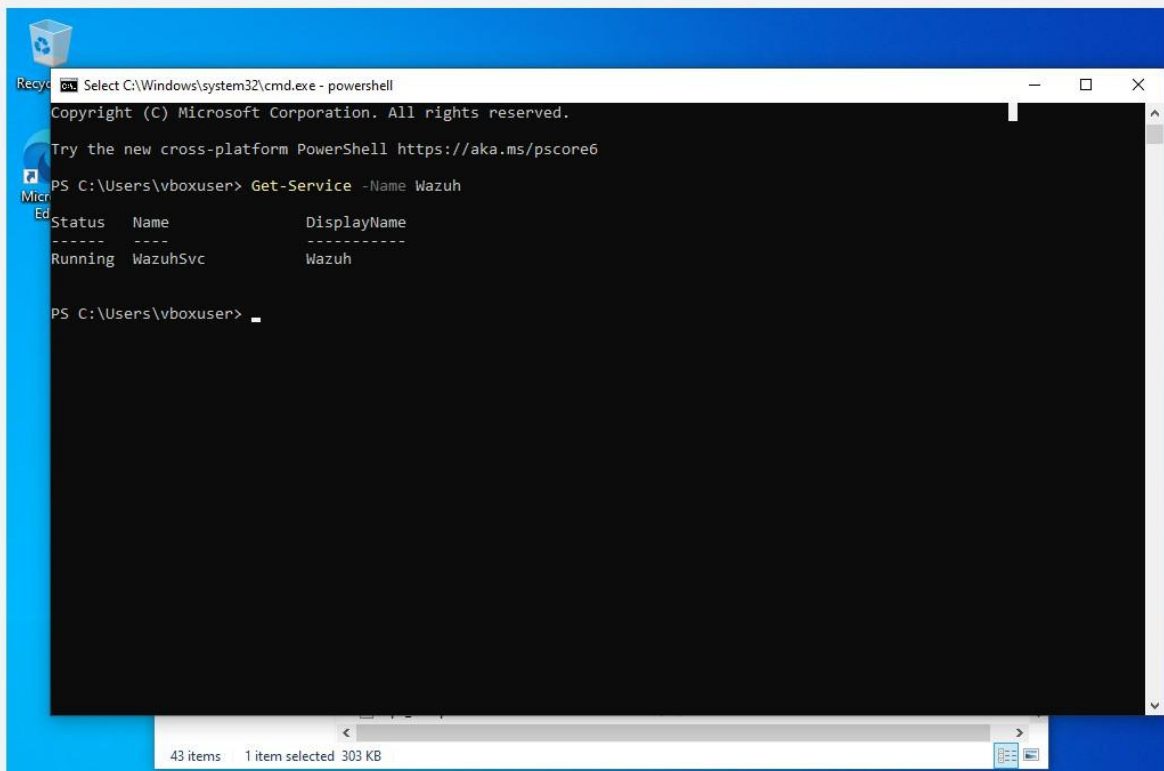


Figure 11 PowerShell showing Wazuh service status as running on the Windows endpoint.

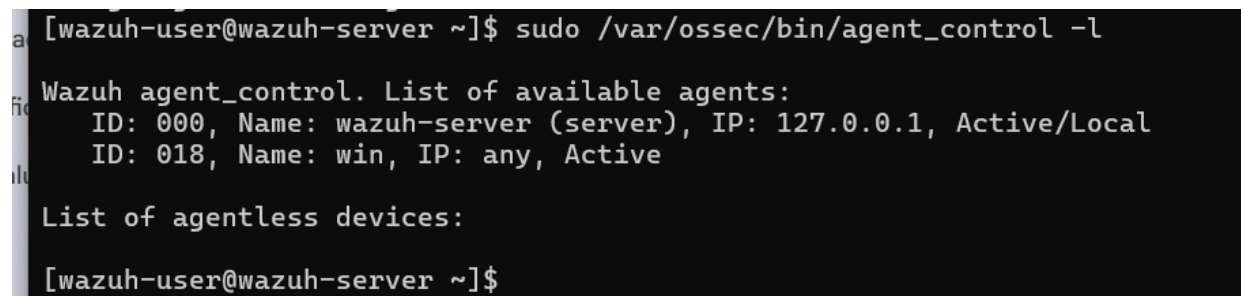


Figure 12 Wazuh agent control listing active agents.

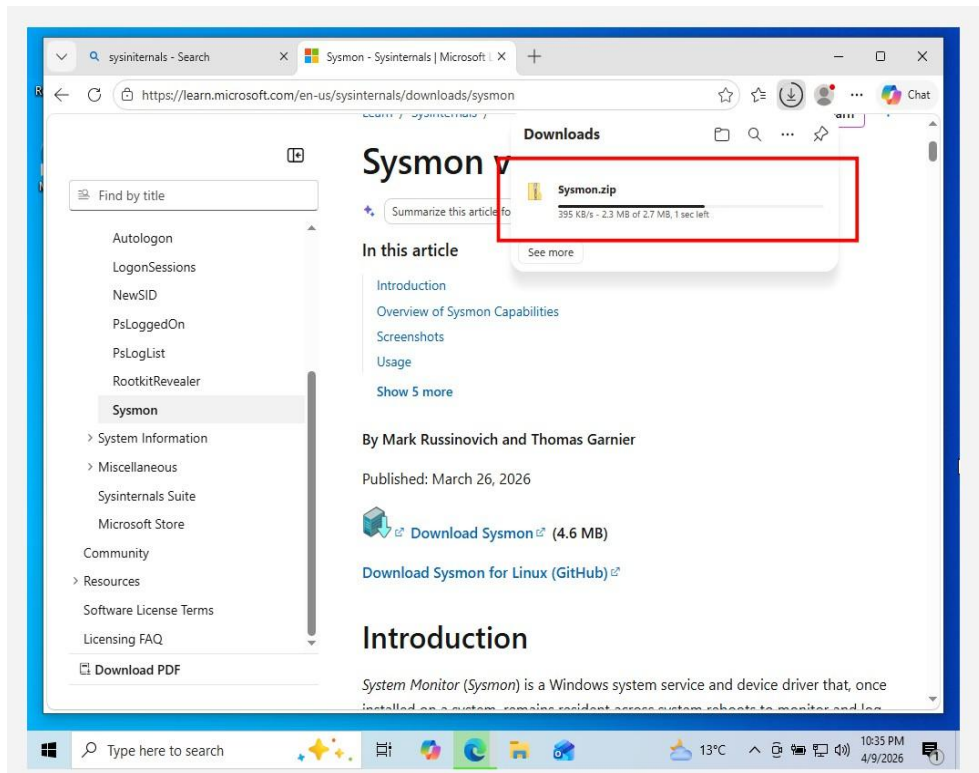


Figure 13 Downloading Sysmon.zip file

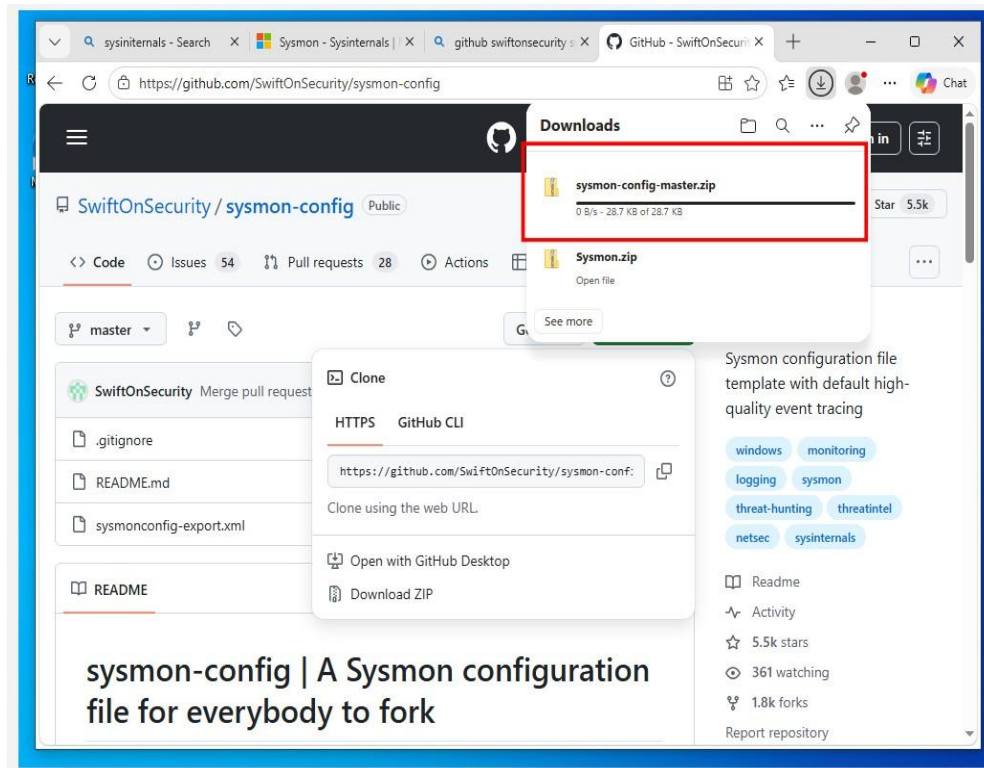


Figure 14 Downloading sysmon-config-master.zip file

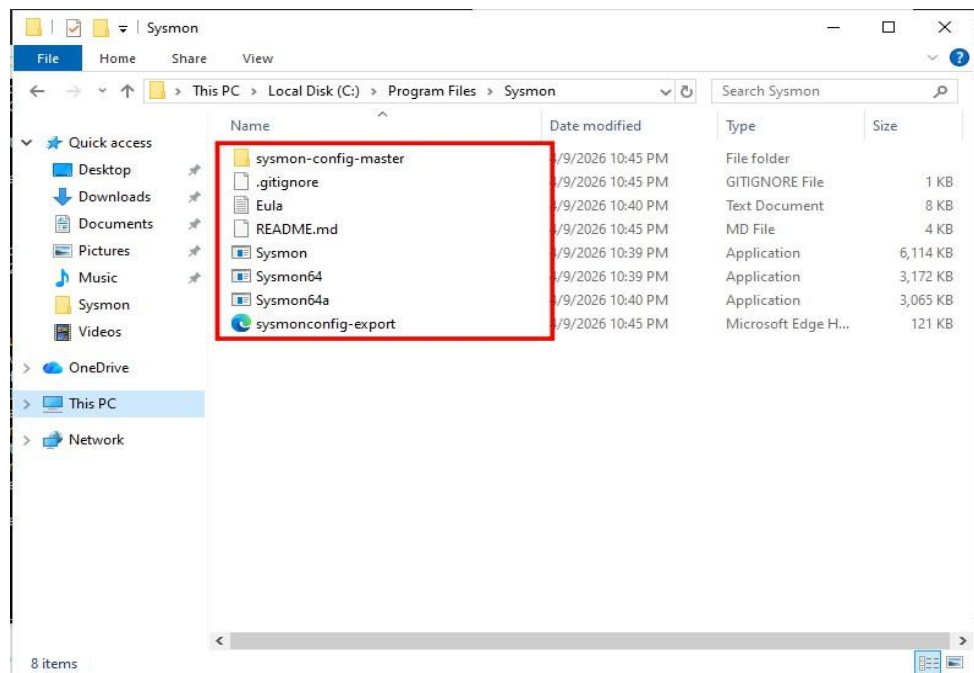


Figure 15 Sysmon configuration files and related applications in the Sysmon folder.

```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>cd c:\program files\sysmon

c:\Program Files\Sysmon>sysmon64.exe -accepteula -i sysmonconfig-export.xml

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
```

Figure 16 Sysmon service started

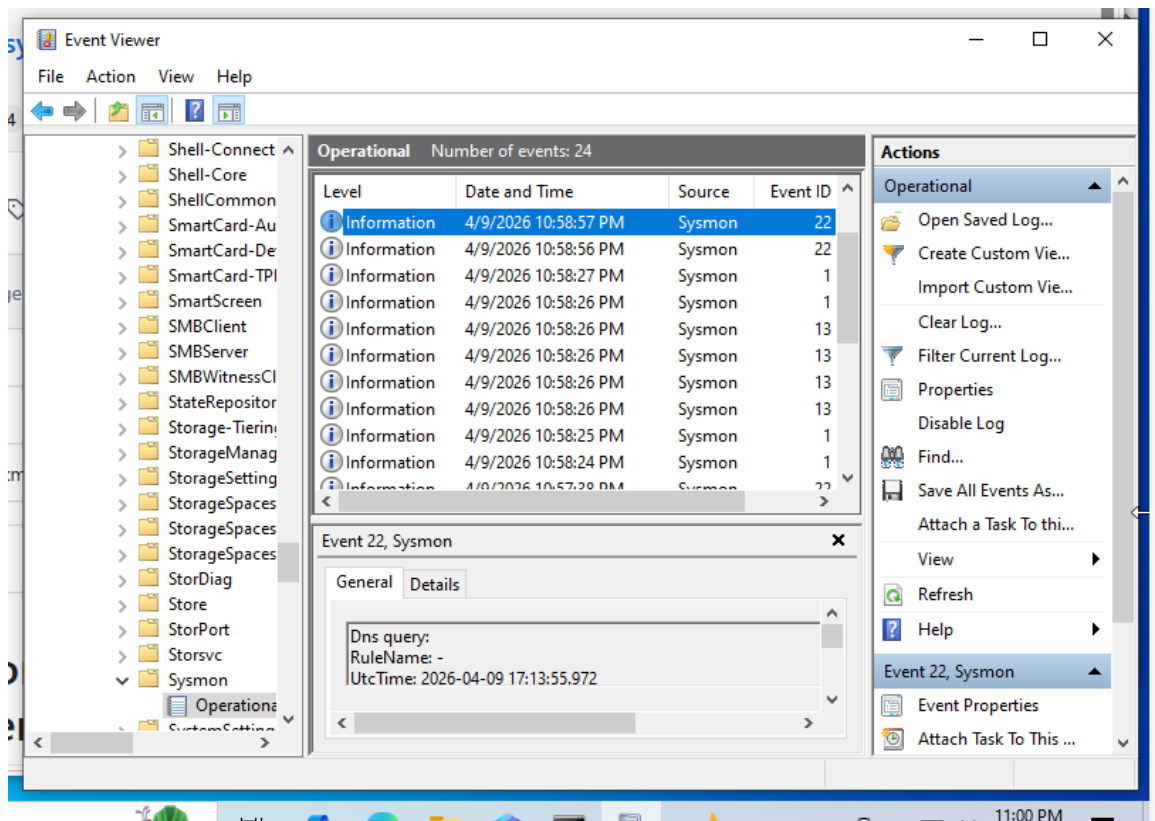


Figure 17 Sysmon event logs in Windows Event Viewer

```
PS C:\Users\vboxuser> Get-Content "C:\Program Files (x86)\ossec-agent\ossec.conf" | Select-String -Pattern
"Sysmon/Operational" -Context 1,2

<localfile>
> <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

Figure 18 PowerShell command extracting Sysmon log configuration from OSSEC